

# WORKING NOTES: IMPROVED REGULARITY FOR A NONLOCAL TWO-PHASE DEAD-CORE PROBLEM DRIVEN BY THE FRACTIONAL $p$ -LAPLACIAN

HIKMATULLO

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## 1. INTRODUCTION (DRAFT)

Dead-core phenomena arise when a diffusing reactant is consumed by a sufficiently strong absorption: the reactant vanishes identically on an interior region, the *dead core*, on whose boundary the solution touches zero together with its gradient. The defining mathematical feature is a sublinear, non-Lipschitz absorption  $u^\gamma$  with  $0 < \gamma < 1$  (local) or  $0 < \gamma < p - 1$  (the  $p$ -Laplacian scaling): it is precisely the failure of Lipschitz continuity at  $u = 0$  that permits a solution to reach zero on a set of positive measure rather than at isolated points. We study the *two-phase, nonlocal* model

$$(1) \quad (-\Delta_p)^s u = u_+^\gamma - u_-^\gamma \quad \text{in } B_1, \quad 0 < \gamma < p - 1, \quad s \in (0, 1), \quad p \geq 2,$$

where  $(-\Delta_p)^s$  is the fractional  $p$ -Laplacian. Two structural choices distinguish (1) from the classical literature. First, the problem is *nonlocal*: the value of  $(-\Delta_p)^s u(x)$  depends on  $u$  over all of  $\mathbb{R}^n$ , so the maximum-principle and minimum-of-the-energy arguments that drive the local one-phase theory are unavailable — in the nonlocal setting a comparison principle requires sign information on the entire complement of  $B_1$ . Second, the problem is *two-phase*:  $u$  is not sign-constrained, so branching points are not local extrema and the one-phase tools fail for a second, independent reason. The one-phase signed problem is in fact obstructed nonlocally by the strong minimum principle, which makes the two-phase formulation the natural setting.

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*Date:* June 15, 2026 — lab notebook, not a draft.

Our main result is a sharp growth estimate at branching points. At a point  $x_0$  with  $u(x_0) = |Du(x_0)| = 0$ ,

$$(2) \quad |u(x)| \leq C \|u\|_{L^\infty(\mathbb{R}^n)} |x - x_0|^\alpha, \quad \alpha = \frac{sp}{p-1-\gamma},$$

an exponent strictly larger than the optimal regularity  $\min(1, \frac{sp}{p-1})$  of  $(s, p)$ -harmonic functions, since  $\gamma > 0$ : solutions are *more* regular at branching points than the homogeneous theory predicts. Passing to the limit  $s \nearrow 1$  yields the analogous estimate for the local two-phase  $p$ -dead-core problem  $\Delta_p u = u_+^\gamma - u_-^\gamma$  with exponent  $\frac{p}{p-1-\gamma}$ , which appears to be new even locally — the existing  $p$ -dead-core results (da Silva–Salort, da Silva–Rossi–Salort, Teixeira) are one-phase.

The method adapts the blow-up/Liouville strategy of dos Prazeres–Teymurazyan–Urbano [1] (the case  $p = 2$ ) and its quasilinear local counterpart in Correa–dos Prazeres [4]. One rescales according to the natural exponent  $\alpha$ , shows the absorption term vanishes along the blow-up, and classifies the limit by a Liouville-type theorem. The new mechanism for  $p \neq 2$  is that the  $(p-1)$ -homogeneity of  $(-\Delta_p)^s$  replaces the linearity used in [1]: it is what forces the right-hand side to vanish in the limit (Proposition 5.1), playing the role that differentiability of the nonlinearity at 0 plays in the fully nonlinear case. The principal difficulties are genuinely nonlocal: controlling the far-field tail of the blow-up family uniformly (§6), and proving the Liouville theorem without the linear structure (§7); the interior  $C^{1,\alpha_0}$  input is the recent estimate of Giovagnoli–Jesus–Silvestre [3].

*Note: Intro drafted 2026-06-15. Keep the exponent claim in (2) phrased against “optimal regularity of  $(s, p)$ -harmonic functions”; the precise relation of  $\alpha$  to  $1+\alpha_0$  is the open bookkeeping point flagged in §8.*

## 2. THE PROBLEM AND THE PROGRAM

We study the two-phase nonlocal dead-core problem

$$(3) \quad (-\Delta_p)^s u = u_+^\gamma - u_-^\gamma \quad \text{in } B_1,$$

with  $0 < \gamma < p-1$ ,  $s \in (0, 1)$ ,  $p \geq 2$ , where

$$(-\Delta_p)^s u(x) := c_{n,s,p} \text{P.V.} \int_{\mathbb{R}^n} \frac{|(u(x) - u(y))|^{p-2} (u(x) - u(y))}{|x - y|^{n+sp}} dy.$$

The goal is the analogue of the Prazeres–Teymurazyan–Urbano (PTU) result [1]: at free boundary points  $x_0$  with  $u(x_0) = |Du(x_0)| = 0$ ,

$$|u(x)| \leq C \|u\|_{L^\infty(\mathbb{R}^n)} |x - x_0|^\alpha, \quad \alpha := \frac{sp}{p-1-\gamma},$$

which exceeds the optimal regularity  $\min(1, \frac{sp}{p-1})$  of  $(s, p)$ -harmonic functions, since  $\gamma > 0$ .

**Program (the PTU machine, transplanted).**

§4 Scaling exponent (Gear 2). *Survives.*

§5 Blow-up limit equation via  $(p-1)$ -homogeneity (replaces linearization). *Survives, new mechanism.*

§6 Constraint region in  $(s, p, \gamma)$  (tail integrability,  $\beta < 1$ , regularity ceiling).

§7 Liouville theorem for entire  $(s, p)$ -harmonic functions with controlled  $C^{1,\alpha_0}$  growth. *The load-bearing wall; likely the main new lemma.*

§8 Target theorems: growth at the free boundary, gradient estimate, Liouville with growth, the limit  $s \nearrow 1$  (two-phase local  $p$ -dead-core: new even locally).

*Note: One-phase signed dead cores are obstructed nonlocally by the strong minimum principle; the two-phase formulation (3) is the natural (and essentially only) nontrivial setting. Use this in the introduction.*

### 3. NOTATION AND STANDING ASSUMPTIONS

For  $\alpha \in (0, 1]$  we use the seminorms  $[\cdot]_{C^\alpha(B_R)}$  and  $[\cdot]_{C^{1+\alpha}(B_R)}$  as in [1, Sec. 2]. The nonlocal tail adapted to  $(-\Delta_p)^s$  is

$$\text{Tail}_{p-1,sp}(u; x_0, R) := \left( R^{sp} \int_{\mathbb{R}^n \setminus B_R(x_0)} \frac{|u(y)|^{p-1}}{|y - x_0|^{n+sp}} dy \right)^{\frac{1}{p-1}},$$

replacing the space  $L_s^1(\mathbb{R}^n)$  of the case  $p = 2$ . **TODO:** Fix the notion of solution for the first pass: weak solutions  $u \in W_{loc}^{s,p} \cap L^\infty(\mathbb{R}^n)$ , with the viscosity equivalence (Korvenpää–Kuusi–Lindgren) cited later. Existence plumbing deferred to the final write-up.

### 4. GEAR 2: THE SCALING EXPONENT

**Proposition 4.1** (Scaling). *Let  $u$  solve (3) and, for  $r > 0$  and  $\alpha > 0$ , set  $u_r(x) := u(rx)/r^\alpha$ . Then*

$$(-\Delta_p)^s u_r(x) = r^{sp-\alpha(p-1)} ((-\Delta_p)^s u)(rx),$$

and  $u_r$  solves (3) (in  $B_{1/r}$ ) if and only if

$$sp - \alpha(p-1) + \alpha\gamma = 0, \quad \text{i.e.} \quad \boxed{\alpha = \frac{sp}{p-1-\gamma}}.$$

*Proof.* Write  $J_p(t) := |t|^{p-2}t$  and note the  $(p-1)$ -homogeneity  $J_p(\lambda t) = \lambda^{p-1}J_p(t)$  for  $\lambda > 0$ . Fix  $x \in B_{1/r}$ . Since  $u_r(x) - u_r(y) = r^{-\alpha}(u(rx) - u(ry))$ ,

$$(-\Delta_p)^s u_r(x) = c_{n,s,p} r^{-\alpha(p-1)} \text{P.V.} \int_{\mathbb{R}^n} \frac{J_p(u(rx) - u(ry))}{|x - y|^{n+sp}} dy.$$

Change variables  $z = ry$ , so  $dy = r^{-n}dz$  and  $|x - y|^{-(n+sp)} = r^{n+sp} |rx - z|^{-(n+sp)}$ ; the kernel and the volume element together contribute  $r^{sp}$ , and we obtain

$$(4) \quad (-\Delta_p)^s u_r(x) = r^{sp-\alpha(p-1)} c_{n,s,p} \text{P.V.} \int_{\mathbb{R}^n} \frac{J_p(u(rx) - u(z))}{|rx - z|^{n+sp}} dz = r^{sp-\alpha(p-1)} ((-\Delta_p)^s u)(rx).$$

Now use the equation. Since  $r^{-\alpha} > 0$ , positive and negative parts scale as  $(u_r)_\pm(x) = r^{-\alpha}u_\pm(rx)$ , whence

$$(u_r)_+^\gamma(x) - (u_r)_-^\gamma(x) = r^{-\alpha\gamma} (u_+^\gamma - u_-^\gamma)(rx).$$

Combining this with (4) and  $(-\Delta_p)^s u(rx) = (u_+^\gamma - u_-^\gamma)(rx)$  for  $rx \in B_1$ ,

$$(-\Delta_p)^s u_r(x) = r^{sp-\alpha(p-1)+\alpha\gamma} [(u_r)_+^\gamma - (u_r)_-^\gamma](x), \quad x \in B_{1/r}.$$

Hence  $u_r$  solves (3) in  $B_{1/r}$  if and only if  $sp - \alpha(p-1) + \alpha\gamma = 0$ , i.e.  $\alpha = \frac{sp}{p-1-\gamma}$  (recall  $\gamma < p-1$ , so the denominator is positive).

The same identity holds for weak solutions: for  $\varphi \in C_c^\infty(B_{1/r})$ , the change of variables  $(x, y) \mapsto (rx, ry)$  in the bilinear form

$$\mathcal{E}(u, \varphi) := \frac{c_{n,s,p}}{2} \iint_{\mathbb{R}^n \times \mathbb{R}^n} \frac{J_p(u(x) - u(y))(\varphi(x) - \varphi(y))}{|x - y|^{n+sp}} dx dy$$

gives  $\mathcal{E}(u_r, \varphi) = r^{sp-\alpha(p-1)-n} \mathcal{E}(u, \varphi(\cdot/r))$ , and the right-hand side scales by the matching factor  $r^{-\alpha\gamma-n}$  after the same substitution; the exponent balance is identical.

*Consistency checks.* For  $p = 2$ :  $\alpha = \frac{2s}{1-\gamma}$ , the PTU exponent [1]. For  $s \nearrow 1$  (with  $p$  fixed):  $\alpha \rightarrow \frac{p}{p-1-\gamma}$ , the local dead-core exponent of da Silva–Salort. Finally, for  $\gamma > 0$  one has  $\alpha = \frac{sp}{p-1-\gamma} > \frac{sp}{p-1}$ , the optimal exponent of  $(s, p)$ -harmonic functions, as claimed.  $\square$

*Remark 4.2* (Homogeneity in the dependent variable). Alongside (4) we record, for  $\lambda > 0$ ,  $(-\Delta_p)^s(\lambda u) = \lambda^{p-1}(-\Delta_p)^s u$ , which is the identity used in the blow-up argument of §5.

*Remark 4.3.*  $\alpha > \frac{sp}{p-1}$  for every  $\gamma > 0$ : the target exponent strictly beats the optimal regularity of the homogeneous equation. This is the “better than Schauder” punchline of the paper.

## 5. GEAR 4': THE BLOW-UP LIMIT EQUATION VIA HOMOGENEITY

In PTU, linearity makes incremental quotients  $s$ -harmonic. For  $p \neq 2$  this fails, but  $(p-1)$ -homogeneity of  $(-\Delta_p)^s$  makes linearization unnecessary: the absorption term dies in the blow-up.

**Proposition 5.1** (Vanishing right-hand side along the blow-up). *Let  $u_k$  be normalized solutions of (3), let  $r_k \rightarrow 0$ ,  $\theta_k \rightarrow \infty$ , and set*

$$v_k(x) := \frac{u_k(r_k x)}{\theta_k r_k^\alpha}, \quad \alpha = \frac{sp}{p-1-\gamma}.$$

*Then*

$$(-\Delta_p)^s v_k(x) = \theta_k^{\gamma-(p-1)} \left[ (v_k)_+^\gamma(x) - (v_k)_-^\gamma(x) \right],$$

*and since  $\gamma < p-1$ , the right-hand side tends to 0 locally uniformly whenever  $v_k$  is locally bounded. Consequently any local uniform limit  $v_0$  is an entire  $(s, p)$ -harmonic function:  $(-\Delta_p)^s v_0 = 0$  in  $\mathbb{R}^n$ .*

*Proof.* Write  $\lambda_k := (\theta_k r_k^\alpha)^{-1} > 0$ , so that  $v_k(x) = \lambda_k u_k(r_k x)$ . We combine two scalings.

*Spatial scaling.* For the dilation  $x \mapsto r_k x$ , the change of variables of Proposition 4.1 (with the multiplicative homogeneity  $(-\Delta_p)^s(\lambda w) = |\lambda|^{p-2} \lambda (-\Delta_p)^s w = \lambda^{p-1} (-\Delta_p)^s w$  of Remark 4.2 absorbing the prefactor) gives the order- $sp$  identity

$$(-\Delta_p)^s [u_k(r_k \cdot)](x) = r_k^{sp} ((-\Delta_p)^s u_k)(r_k x).$$

*Amplitude scaling.* By  $(p-1)$ -homogeneity in the dependent variable,

$$(-\Delta_p)^s v_k(x) = (-\Delta_p)^s [\lambda_k u_k(r_k \cdot)](x) = \lambda_k^{p-1} r_k^{sp} ((-\Delta_p)^s u_k)(r_k x).$$

*Use the equation.* For  $r_k x \in B_1$ ,  $(-\Delta_p)^s u_k(r_k x) = u_k(r_k x)_+^\gamma - u_k(r_k x)_-^\gamma$ . Since  $\lambda_k > 0$ ,  $(v_k)_\pm(x) = \lambda_k u_k(r_k x)_\pm$ , whence  $u_k(r_k x)_\pm^\gamma = \lambda_k^{-\gamma} (v_k)_\pm^\gamma(x)$  and

$$(-\Delta_p)^s v_k(x) = \lambda_k^{p-1-\gamma} r_k^{sp} \left[ (v_k)_+^\gamma - (v_k)_-^\gamma \right](x).$$

*The exponents collapse.* By definition of  $\lambda_k$ ,  $\lambda_k^{p-1-\gamma} r_k^{sp} = \theta_k^{-(p-1-\gamma)} r_k^{sp-\alpha(p-1-\gamma)}$ , and  $\alpha(p-1-\gamma) = sp$  exactly (this is the defining property of  $\alpha$ ), so the power of  $r_k$  vanishes and

$$(-\Delta_p)^s v_k(x) = \theta_k^{\gamma-(p-1)} \left[ (v_k)_+^\gamma - (v_k)_-^\gamma \right](x), \quad x \in B_{1/r_k}.$$

Since  $0 < \gamma < p-1$ , the exponent  $\gamma-(p-1) < 0$  and  $\theta_k \rightarrow \infty$ , so the prefactor  $\rightarrow 0$ ; if  $v_k$  is locally bounded the bracket is locally bounded and the right-hand side  $\rightarrow 0$  locally uniformly. Passing to the limit (stability of weak/viscosity solutions of  $(-\Delta_p)^s$  under local uniform convergence with tail control, see the discussion below) yields  $(-\Delta_p)^s v_0 = 0$  in  $\mathbb{R}^n$ .  $\square$

*Note: The cancellation is the whole mechanism: spatial scaling supplies  $r_k^{sp}$ , amplitude scaling supplies  $(\theta_k r_k^\alpha)^{-(p-1-\gamma)}$ , and the choice  $\alpha = \frac{sp}{p-1-\gamma}$  is precisely what makes the two powers of  $r_k$  cancel, leaving the pure damping factor  $\theta_k^{-(p-1-\gamma)} \rightarrow 0$ . This is what we mean by “homogeneity replaces linearization”. The only analytic input still to pin down is the stability lemma for  $(-\Delta_p)^s$  (Brasco–Lindgren / Korvenpää–Kuusi–Lindgren), the  $p \neq 2$  replacement for [2, Lemma 5]; it carries the same global tail-convergence hypothesis whose verification is the subject of §6. Note: This parallels PTU §6 (fully nonlinear case), where  $\delta^{-1}F(\delta M) \rightarrow DF(0)M$ ; here homogeneity plays the role of differentiability of  $F$  at 0 — which  $(-\Delta_p)^s$  does not have. Worth a remark in the paper.*

## 6. GEAR 3 BOOKKEEPING: THE ADMISSIBLE REGION IN $(s, p, \gamma)$

Three constraints must hold simultaneously. Set  $\alpha = \frac{sp}{p-1-\gamma}$  and  $\beta := \alpha - 1$  (the  $\nu = 1$  case).

**Lemma 6.1** (Tail integrability). *The growth  $|v_0(x)| \lesssim |x|^\alpha$  at infinity is compatible with  $\text{Tail}_{p-1,sp}(v_0; 0, R) < \infty$  for all  $R$  if and only if  $\alpha(p-1) < sp + p - 1$ , i.e.*

$$\gamma < (p-1) - \frac{sp}{1 + sp/(p-1)} = (p-1) \left( 1 - \frac{sp}{p-1+sp} \right) = \frac{(p-1)^2}{p-1+sp}$$

*Verification (2026-06-10; constants to be rechecked).* We record the bookkeeping for the blow-up family  $v_k(x) = \frac{u_k(r_k x)}{\theta_k r_k^\alpha}$ ,  $\|u_k\|_{L^\infty(\mathbb{R}^n)} \leq M$ , under the seminorm normalization  $\theta_k = \theta_k(r_k)$  with  $\theta_k(r') := \sup_{r' < r < 1/2} r^{1+\alpha'-\alpha} [u_k]_{C^{1+\alpha'}(B_r)}$ .

*Step 1: uniform tails for  $v_k$  itself are impossible.* The interior information gives the growth bound  $|v_k(y)| \leq C|y|^\alpha$  for  $1 \leq |y| \leq R_k^* := \frac{1}{2r_k}$  (gradient-anchor trick of [4]), while globally  $|v_k| \leq M(\theta_k r_k^\alpha)^{-1}$ . The two bounds cross at  $\rho_k = (M/C\theta_k)^{1/\alpha} r_k^{-1}$ . Splitting the tail there and using  $\alpha(p-1) - sp = \alpha\gamma > 0$ ,

$$\text{Tail}_{p-1,sp}^{p-1}(v_k; 0, R) \leq c R^{\alpha(p-1)} + c R^{sp} \rho_k^{\alpha\gamma} = c R^{\alpha(p-1)} + c R^{sp} (\theta_k r_k^\alpha)^{-\gamma},$$

with both the growth region and the capped region contributing comparably. But the normalization forces  $\theta_k \leq [u_k]_{C^{1+\alpha'}(B_{1/2})} r_k^{1+\alpha'-\alpha} \leq 2C r_k^{1+\alpha'-\alpha}$ , i.e.  $\theta_k r_k^\alpha \leq 2C r_k^{1+\alpha'} \rightarrow 0$ ; hence  $\sup_k \text{Tail}_{p-1,sp}(v_k; 0, R) = \infty$ . **Conclusion: the  $C^{1,\alpha_0}$  estimate can never be applied to  $v_k$  directly; one must work one order down.** This settles the “decide” in the old TODO: the incremental-quotient step (PTU (4.11)) is *not* optional.

*Step 2: incremental quotients give exactly the boxed condition.* For  $w_k^h := \frac{v_k(\cdot+h) - v_k(\cdot)}{|h|}$ ,  $0 < |h| \leq 1$ , the gradient bound yields  $|w_k^h(y)| \leq C(1+|y|)^{\alpha-1}$  for  $|y| \leq R_k^*$ , and now the growth-region tail integral

$$R^{sp} \int_{|y|>R} |y|^{(\alpha-1)(p-1)-n-sp} dy$$

converges *uniformly in  $k$*  if and only if  $(\alpha-1)(p-1) < sp$ , which upon substituting  $\alpha = \frac{sp}{p-1-\gamma}$  is precisely

$$\gamma < \frac{(p-1)^2}{p-1+sp}.$$

Sanity check at  $p = 2$ :  $\gamma < \frac{1}{1+2s}$ , matching PTU, and  $\gamma < 1/3$  uniformly for  $s \in (\frac{1}{2}, 1)$ . ✓

*Step 3: the residual far-field gap.* Beyond  $R_k^*$  the cap gives the contribution  $c R^{sp} |h|^{-(p-1)} \theta_k^{-(p-1)} r_k^{-\alpha\gamma}$ , which is bounded iff  $\theta_k \geq c r_k^{-\alpha\gamma/(p-1)}$  — a *lower* bound on  $\theta_k$  that the contradiction hypothesis does not obviously provide ( $\theta_k \rightarrow \infty$ , but at an unknown rate relative to  $r_k$ ). **TODO: (Checked**

2026-06-10.) PTU §4 does *not* obviously provide this bound: their (4.11) rests on (4.8), valid only for  $|x| \leq \frac{1}{2r_k}$ , while the limit passage (4.12) needs uniform  $L_s^1$  control on all of  $\mathbb{R}^n$ ; the

outer region contributes  $\sim \theta_k^{-1} r_k^{-\frac{2s\gamma}{1-\gamma}}$  — the same lower bound, apparently unaddressed. So the far-field issue is not an artifact of  $p \neq 2$ : it is present (implicitly resolved, or genuinely open) in the template itself. **Priority question for R.T.** Candidate fixes: dichotomy  $\theta_k \geq r_k^{-\delta}$ ; cutoff with error estimate. (Checked: weakening the stability lemma is *not* an option — [2] Lemma 4.3 requires global  $L^1(\omega)$  convergence, and their Remark 4.4 exhibits a counterexample of exactly this far-field-mass type.)  $\square$

**Lemma 6.2** (Order of the jet:  $\beta < 1$ ). *The Liouville theorem with  $\nu = 1$  requires  $\beta = \frac{sp}{p-1-\gamma} - 1 < 1$ , i.e.*

$$\gamma < (p-1) - \frac{sp}{2}$$

which also forces  $sp < 2(p-1)$ .

**Lemma 6.3** (Regularity ceiling). *The compactness step requires a uniform interior estimate one derivative above the vanishing jet. With  $\nu = 1$  this means local  $C^{1,\alpha_0}$  estimates for normalized solutions of  $(-\Delta_p)^s u = f \in L^\infty$ , currently available for*

$$p \in \left[ 2, \frac{2}{1-s} \right),$$

together with the constraint  $\alpha < 1 + \alpha_0$ .

*Note: (2026-06-10, after reading [3].) Two structural facts. (i) [3] is stated for  $(-\Delta_p)^s u = 0$ ; by their own remark the extension to  $f \in L^\infty$  is expected when the linearized operator has order  $2 - p(1-s) > 1$ , i.e.  $p < \frac{1}{1-s}$  — which forces  $s > \frac{1}{2}$ , matching PTU. Their stage-3 machinery (Imbert–Silvestre) already takes a right-hand side, and the Lipschitz starting point with data exists (Biswas–Sen, arXiv:2507.09920). Writing this extension out is a candidate lemma of the present paper. (ii) **Warning for Theorem 8.3:** the constants in [3] are explicitly NOT uniform as  $s \nearrow 1$  (their §1.2: the first stage is “purely nonlocal”). The  $s \nearrow 1$  route to the local two-phase theorem is blocked through this estimate; the local theorem should instead be proved directly (classical  $C^{1,\alpha}$  theory for the  $p$ -Laplacian, run through the scheme of [4] §6 with right-hand side  $u_+^\gamma - u_-^\gamma$ ). **TODO:** (i) Track down the  $C^{1,\alpha_0}$  paper (cited as [28] in arXiv:2509.15988) and record the exact statement: dependence of  $\alpha_0$  and of the constant on  $n, s, p$ , presence of tail terms, and uniformity as  $s \nearrow 1$  (needed for §8, Theorem 8.3). (ii) Intersect the three boxed regions and draw the admissible set; this intersection is the hypothesis of the main theorem, the analogue of  $\gamma \in (0, 1/3)$ ,  $s \in (1/2, 1)$  in PTU. (iii) Decide whether a weaker fallback (only  $C^{0,\gamma_0}$  estimates,  $\gamma_0 = \min(1, \frac{sp}{p-1})$ , with the free boundary condition  $u(x_0) = 0$  only and  $\alpha \leq \gamma_0$  forced) yields a nonempty parameter range — this would give an unconditional but weaker theorem.*

**Attack notes on the far-field bound (2026-06-10; heuristic, exponents to be redone slowly on paper).** Write the needed bound as

$$(N) \quad \theta_k^{p-1} r_k^{\alpha\gamma} \geq c > 0.$$

(a) *Compatibility constraint on the working seminorm order.* The device forces  $\theta_k \leq 2C r_k^{1+\alpha'-\alpha}$ , so (N) is consistent with the worst case only if  $r_k^{1+\alpha'-\alpha} \gtrsim r_k^{-\alpha\gamma/(p-1)}$  for small  $r$ , i.e.

$$1 + \alpha' \leq \alpha \frac{p-1-\gamma}{p-1} = \frac{sp}{p-1}, \quad \text{equivalently} \quad \alpha' \leq \frac{sp - (p-1)}{p-1},$$

which requires  $s > 1 - \frac{1}{p}$ . At  $p = 2$  this reads  $s > \frac{1}{2}$  — precisely the PTU range; plausibly the structural reason for it.

(b) *Far field as a slowly varying datum — the likely implicit fix in PTU.* For  $x \in B_R$ , split the operator at  $R_k^* = \frac{1}{2r_k}$  and treat the far region as a datum  $F_k(x)$ . Its *size* diverges like  $\theta_k^{-(p-1)} r_k^{-\alpha\gamma}$  when (N) fails — but its *oscillation* on  $B_R$  is much better. At  $p = 2$ : the divergent part of  $F_k$  is constant in  $x$  and increments kill it; the  $x$ -variation is controlled by one extra power of kernel decay,

$$\text{cap}_k(R_k^*)^{-2s-1} = M\theta_k^{-1}r_k^{2s+1-\alpha} \rightarrow 0 \iff \gamma < \frac{1}{1+2s},$$

*exactly the tail condition.* So for  $p = 2$  the limit equation for increments survives without any lower bound on  $\theta_k$ : stability should be applied to the *localized* function with the far field carried as a datum of vanishing oscillation, not via global  $L_s^1$  convergence. **TODO: Make this rigorous at  $p = 2$  first; it would repair PTU (4.12) and is publishable as a remark/appendix.**

(c) *The  $p \neq 2$  version.* Truncate:  $\tilde{v}_k = v_k \chi_{B_{R_k^*/2}}$  solves the equation in  $B_R$  up to an error  $G_k(x) = c \int_{|y| > R_k^*/2} [(v_k(x) - v_k(y))^{p-2} (v_k(x) - v_k(y)) - |v_k(x)|^{p-2} v_k(x)] |x - y|^{-n-sp} dy$ . Decompose  $G_k = a_k + g_k$ ,  $a_k$  constant. Two estimates:

$$\|\nabla g_k\|_{L^\infty(B_R)} \lesssim \underbrace{\text{cap}_k^{p-2}(R_k^*)^{-sp}}_{\rightarrow 0 \text{ since } sp - \alpha(p-2) = \alpha(1-\gamma) > 0} + \underbrace{\text{cap}_k^{p-1}(R_k^*)^{-sp-1}}_{\rightarrow 0 \iff \gamma < \frac{p-1}{1+sp}}.$$

The new admissibility condition

$$\boxed{\gamma < \frac{p-1}{1+sp}}$$

coincides with the tail condition  $\gamma < \frac{(p-1)^2}{p-1+sp}$  at  $p = 2$  and is *stricter* for  $p > 2$ ; if (b)/(c) are the right mechanism, this replaces (or joins) Lemma 6.1 in the hypotheses. The divergent constant  $a_k$ : for  $p = 2$  increments kill it; for  $p \neq 2$  **TODO: decide what kills  $a_k$  — a posteriori boundedness (if  $a_k \rightarrow \infty$ , interior estimates on the now-tail-finite  $\tilde{v}_k$  force a contradiction in the equation), or a smarter truncation. This is the crux for  $p \neq 2$ .**

(c') *Proposed repair at  $p = 2$  (2026-06-10; complete in outline, every step to be checked).* Let  $w_k = v_k(\cdot + h) - v_k(\cdot)$  as in PTU (4.11)–(4.12),  $\rho_k := \frac{1}{4r_k}$ , and truncate:  $\tilde{w}_k := w_k \chi_{B_{\rho_k}}$ . For  $x \in B_R$ ,

$$(-\Delta)^s \tilde{w}_k(x) = f_k(x) + H_k(x), \quad H_k(x) := c \int_{|y| > \rho_k} w_k(y) |x - y|^{-n-2s} dy,$$

with  $f_k \rightarrow 0$  locally uniformly (PTU (4.12)). Then:

- (1) *Oscillation of  $H_k$  vanishes:*  $|\nabla H_k| \lesssim C_h \rho_k^{\alpha-2s-2} + \theta_k^{-1} r_k^{2s+1-\alpha} \rightarrow 0$ , using the growth bound in the annulus and the cap beyond, with  $2s + 1 - \alpha > 0 \iff \gamma < \frac{1}{1+2s}$ . So  $H_k = a_k + o(1)$  with  $a_k$  constant.
- (2)  *$a_k$  is bounded a posteriori:*  $\tilde{w}_k$  has uniform local  $C^{2s+\gamma'}$  bounds and uniform  $L_s^1$  norm (growth  $|y|^{\alpha-1}$ , integrable by the tail condition), so  $|(-\Delta)^s \tilde{w}_k(x_0)| \leq A$  uniformly at a fixed point; since  $f_k \rightarrow 0$ ,  $|a_k| \leq A + o(1)$ . (No lower bound on  $\theta_k$  ever needed.)
- (3) *Limit equation with a constant:* dominated convergence gives  $\tilde{w}_k \rightarrow w_0$  in  $L_s^1$ ; CS stability now applies and yields  $(-\Delta)^s (v_0(\cdot + h) - v_0) = a(h)$  in  $B_R$ , all  $R$ , where  $a(h) := \lim a_k$  (subsequence).
- (4)  *$a(h)$  is linear in  $h$ :* the cocycle identity  $w^{h_1+h_2} = w^{h_1}(\cdot + h_2) + w^{h_2}$  and translation invariance give  $a(h_1 + h_2) = a(h_1) + a(h_2)$ ; with measurability,  $a(h) = b \cdot h$ .
- (5) *Second differences kill  $b$ :*  $\Delta_h^2 v_0 = w^h(\cdot + h) - w^h$  satisfies  $(-\Delta)^s \Delta_h^2 v_0 = a(h) - a(h) = 0$  entire, with growth  $|\Delta_h^2 v_0(x)| \leq C_h(1+|x|)^{\alpha-2}$ . PTU Theorem 3.1 ( $\nu = 1, \beta = \alpha - 2 < 1$ ) applies:  $\Delta_h^2 v_0$  is affine for every  $h$ , hence  $v_0$  is a polynomial of degree  $\leq 3$ ; growth  $\alpha < 1 + 2s < 3$  cuts it to degree  $\leq 2$ ; the vanishing 2-jet at the branching point ( $\nu = 2$ )



gives  $v_0 \equiv 0$ , contradicting the normalization. *The Liouville input is even weaker than PTU's original use.*

**TODO:** Check (1)–(5) line by line, especially: the uniform local regularity needed in (2) (PTU's  $C^{2s+\gamma}$  bounds suffice?); measurability/continuity of  $a(h)$  in (4); and that the degree- $\leq 3$  deduction in (5) is correct ( $\Delta_h^2 v$  affine  $\forall h \iff v$  cubic — standard, but verify the nonsmooth case). If this holds, it repairs PTU (4.12) *and* removes the need for any far-field lower bound. Then port to  $p \neq 2$ :  $w_k^h$  solves the *linear* equation  $L_{K_k} w_k^h = \text{RHS-difference with the GJS kernel } K_k \simeq |v_k(x) - v_k(y)|^{p-2} |x - y|^{-n-sp}$  (exactly the object of GJS §7); the datum trick is linear-friendly, and the new constraint  $\gamma < \frac{p-1}{1+sp}$  from (c) should be what makes the kernel-datum oscillation vanish. The cocycle/second-difference steps (4)–(5) need rethinking for the vanishing-jet ( $\nu = 1$ ) formulation — but there  $a(h)$  may not even appear, since the CP-style Liouville works with  $v$  itself.

*Remark 6.4* (Verification verdict on (c'), 2026-06-15). A careful pass over steps (1)–(5) at  $p = 2$  finds the argument *sound in outline*, under the single hypothesis  $\gamma < \frac{1}{1+2s}$  (the tail condition). Specifically:

- (1)  $|\nabla H_k|$  splits into an annulus piece  $\lesssim C_h \rho_k^{\alpha-2s-2}$  (with  $\alpha - 2s - 2 < 0$  since  $\alpha < 3 < 2s + 2$  for  $s > \frac{1}{2}$ ) and a cap piece  $\lesssim \theta_k^{-1} r_k^{2s+1-\alpha}$  (with  $2s + 1 - \alpha > 0 \iff \gamma < \frac{1}{1+2s}$ ); both  $\rightarrow 0$ . So  $H_k = a_k + o(1)$  in  $C^1(B_R)$ ,  $a_k$  constant. ✓
- (2) **Not circular.** Decompose  $(-\Delta)^s \tilde{w}_k(x_0) = (\text{local singular part}) + (\text{tail})$ . The local part is bounded by the uniform interior  $[\cdot]_{C^{2s+\gamma}}$  bound on  $w_k$  (PTU (4.8),  $2s + \gamma > 2s$ ); the tail  $\|\tilde{w}_k\|_{L_s^1} = \int_{|y|>1} |w_k(y)| |y|^{-n-2s} dy$  uses *only* the growth bound  $|w_k(y)| \leq C_h |y|^{\alpha-1}$  on the support  $B_{\rho_k}$ , which converges (uniformly in  $k$ ) iff  $\alpha - 1 - 2s < 0$ , again  $\gamma < \frac{1}{1+2s}$ . Neither bound uses  $a_k$  or any lower bound on  $\theta_k$ . Then  $a_k = (-\Delta)^s \tilde{w}_k(x_0) - f_k(x_0) - o(1)$  is bounded a posteriori. ✓ (*This was the flagged “riskiest” step; it holds. The truncation manufactures the very  $L_s^1$  control PTU could not get globally.*)
- (3) Dominated convergence ( $|w_k| \leq C_h(1 + |y|)^{\alpha-1} \in L_s^1$ , pointwise  $\tilde{w}_k(y) \rightarrow w_0(y)$  since eventually  $y \in B_{\rho_k}$ ) gives  $\tilde{w}_k \rightarrow w_0$  in  $L_s^1$  globally; CS stability then legitimately applies (its global hypothesis is now met) and yields  $(-\Delta)^s(v_0(\cdot + h) - v_0) = a(h)$ . ✓
- (4) Cocycle  $w^{h_1+h_2} = w^{h_1}(\cdot + h_2) + w^{h_2}$  plus translation invariance of  $(-\Delta)^s$  gives  $a(h_1+h_2) = a(h_1) + a(h_2)$ ; with measurability (pointwise limit of  $h \mapsto a_k(h)$ , continuous),  $a(h) = b \cdot h$ . ✓ *Minor: extract one diagonal subsequence over a countable dense set of  $h$ , then use continuity, so  $a(\cdot)$  is well defined.*
- (5)  $\Delta_h^2 v_0$  is entire  $s$ -harmonic with growth  $(1 + |x|)^{\alpha-2}$ ,  $\alpha - 2 < 1$ , hence *constant* (sublinear-growth Liouville); thus  $\Delta_h^3 v_0 \equiv 0$ , so  $v_0$  is a quadratic polynomial; the vanishing 2-jet at the branching point forces  $v_0 \equiv 0$ , contradicting the normalization. ✓

**Conclusion:** (c') repairs PTU (4.12) at  $p = 2$  with no lower bound on  $\theta_k$ . The mechanism is truncate  $\rightarrow$  oscillation vanishes  $\rightarrow$  constant bounded a posteriori  $\rightarrow$  stability now legal  $\rightarrow$  second differences annihilate the constant.

**The  $p \neq 2$  obstruction, located precisely.** The datum steps (1)–(3) port (the increment  $w_k^h$  solves the linear  $L_{K_k}$  equation, Remark 7.6, and truncation is linear-friendly). But steps (4)–(5) use *translation invariance* of the operator ( $(-\Delta)^s w^{h_1}(\cdot + h_2) = a(h_1)$ ), and the  $v_k$ -dependent kernel  $K_k$  is *not* translation invariant. So the cocycle/second-difference finish does not transplant. The fix is to drop the additive-constant route altogether and use the vanishing-jet (CP) Liouville on  $v_0$  directly, where  $a(h)$  never appears — consistent with the restatement in Conjecture 7.1. *This is the one item to confirm with R.T.: that the vanishing-jet formulation removes the need for steps (4)–(5) in the nonlinear setting.*



(d) *A warning on pointwise evaluation.* Evaluating  $(-\Delta_p)^s \tilde{v}_k$  classically at a point with only  $C^{1,\alpha'}$  regularity needs (after odd-cancellation of the gradient term in the P.V.) the remainder exponent  $p-1+\alpha'-sp > -\dots$ , i.e.  $\alpha' > sp-(p-1)$  when  $sp > p-1$  — which *clashes* with the upper bound in (a): the window  $sp-(p-1) < \alpha' \leq \frac{sp-(p-1)}{p-1}$  is empty for  $p > 2$  and borderline-empty at  $p = 2$ . **TODO:** This suggests (a) and (d) do not actually bind simultaneously (PTU work at order  $\nu = 2$  where the bookkeeping differs, and weak formulations avoid pointwise evaluation altogether). Sort out which constraints are real in the weak framework — priority for the next session, fresh head needed.

**Synthesis: intersecting the regions (2026-06-15).** Working at order  $\nu = 1$  (the only order available for  $p \neq 2$ , since quadratics are not  $(s, p)$ -harmonic), the explicit constraints are

$$(C1) \text{ tail: } \gamma < \frac{(p-1)^2}{p-1+sp}, \quad (C2) \text{ jet } \beta < 1: \gamma < (p-1) - \frac{sp}{2}, \quad (C4) \text{ datum: } \gamma < \frac{p-1}{1+sp},$$

together with the soft *ceiling* (C3)  $\alpha < 1 + \alpha_0$  and the [3] window  $p \in [2, \frac{2}{1-s})$  (equivalently  $s > 1 - \frac{2}{p}$ ). Three findings from the bookkeeping (all verified symbolically):

- (i) **C4 is the binding explicit constraint for  $p > 2$ .** One has  $C4 \leq C1$  with equality only at  $p = 2$  ( $\frac{C4}{C1} = \frac{p-1+sp}{(p-1)(1+sp)} \leq 1$ ); and C4 is typically below C2 as well (e.g.  $p = 3, s = \frac{3}{4}$ :  $C4 = 0.62 < C2 = 0.88 < C1 = 0.94$ ). So the explicit region is  $\gamma < \min(C2, C4)$  and is *nonempty*.
- (ii) **Restricting to  $\nu = 1$  is not fatal for the new ( $p > 2$ ) range.** The  $\nu = 1$  cap C2 reads  $\gamma < (p-1) - \frac{sp}{2}$ ; at  $s = 1$  this is  $\gamma < \frac{p}{2} - 1$ , which is *positive for every  $p > 2$* . Only at  $p = 2$  does the window pinch to 0 as  $s \nearrow 1$  — which is exactly why PTU needed  $\nu = 2$  there, and exactly the case already covered by [1]. So the unavailability of  $\nu = 2$  costs us nothing where the result is new.
- (iii) **The real binding constraint is the ceiling (C3),  $\alpha < 1 + \alpha_0$ .** Since  $\alpha = \frac{sp}{p-1-\gamma} > 1$  in the relevant range, this needs the [3] exponent to satisfy  $\alpha_0 > \alpha - 1 = \frac{sp-(p-1)+\gamma}{p-1-\gamma}$ . At  $p = 3, s = \frac{3}{4}, \gamma = \frac{1}{2}$  this demands  $\alpha_0 > \frac{1}{2}$  — a *quantitative* lower bound on  $\alpha_0$  that [3] (existence of some  $\alpha_0$ ) does not by itself supply.

**TODO:** Decision point for the meeting. The explicit region  $\gamma < \min(C2, C4)$  is fine; the theorem is conditional on a *quantitative*  $\alpha_0$  from [3] (or a regime where  $\alpha - 1$  is small, i.e.  $\gamma$  small and  $sp$  close to  $p-1$ ). Two honest options: (A) state the main theorem with hypothesis “ $\alpha < 1 + \alpha_0$ ” and carry  $\alpha_0$  as an abstract constant (clean, but the admissible set is then only proved nonempty in the perturbative corner  $sp \downarrow p-1$ ); (B) extract or lower-bound  $\alpha_0(n, s, p)$  from the [3] proof (harder, but yields an explicit region). Ask R.T. which framing he prefers; this is the analogue of PTU’s clean  $\gamma \in (0, 1/3)$ .

## 7. THE LOAD-BEARING WALL: A LIOUVILLE THEOREM

**Conjecture 7.1** (Liouville with vanishing jet,  $\nu = 1$ ; restated 2026-06-15, cap  $\beta < 1$  via increments). *Let  $v \in C_{loc}^1(\mathbb{R}^n)$  be an entire  $(s, p)$ -harmonic function with*

$$v(0) = |Dv(0)| = 0 \quad \text{and} \quad |v(x)| \leq C(1 + |x|)^{1+\beta} \quad \text{in } \mathbb{R}^n,$$

*for some  $0 < \beta < 1$  (equivalently  $\alpha < 2$ , i.e. Lemma 6.2, C2), with the tail condition (C1)  $((1+\beta)(p-1) < sp + (p-1))$ , i.e. Lemma 6.1 in force, and assume  $v$  is nondegenerate ( $Dv \not\equiv 0$  on any open set). Then  $v \equiv 0$ .*

**Remark 7.2** (Why the cap is  $\beta < 1$ , not  $\beta < \alpha_0$  — the route is forced, 2026-06-15). The earlier “ $\beta < \alpha_0$ ” form copied [4]’s *local* Lemma 5, whose proof rescales  $v_l = v(l \cdot)/l^{1+\beta}$  and applies the sharp  $C^{1,\alpha_0}$  estimate to  $v_l$ . **That route is blocked nonlocally:**  $v_l$  has the same growth  $1+\beta$  as

$v$ , and  $\text{Tail}_{p-1,sp}(v_l) < \infty$  would require  $(1+\beta)(p-1) < sp$  — but the target  $1+\beta = \alpha = \frac{sp}{p-1-\gamma}$  gives  $\alpha(p-1) - sp = \frac{\gamma ps}{p-1-\gamma} > 0$  for *every*  $\gamma > 0$ , so the tail diverges; truncating leaves a far-field datum with the same divergent exponent, so [3] cannot be applied to  $v_l$  at all. The route that *does* survive nonlocally is PTU's: pass to increments  $w^h = v(\cdot + h) - v$ , which lower the growth by one order to  $\beta < 1$  and (under C1) have finite tail. This is why the correct cap is  $\beta < 1$  ( $\alpha < 2$ ), exactly the cap of PTU Theorem 3.1, and why the sharp exponent  $\alpha_0$  is needed only for compactness (any  $\alpha_0 > 0$ ), not quantitatively. *This supersedes synthesis finding (iii) and the “need a quantitative  $\alpha_0$ ” worry: under the increment route the binding constraints are C1 and C2, both explicit.* The price is instead the **nondegeneracy** hypothesis, which controls the  $v$ -dependent kernel  $K_v^h$  (Proposition 7.5, Remark 7.6).

**Lemma 7.3** (Jet-to-growth; [4] Lemma 6, transplanted). *Let  $u \in C^{1+\alpha'}(B_1)$  with  $0 < \alpha' < 1$ , where  $[u]_{C^{1+\alpha'}} := [Du]_{C^{0,\alpha'}}$ . If  $u(0) = |Du(0)| = 0$  and*

$$\sup_{0 < r < 1/2} r^{\alpha' - \beta} [u]_{C^{1+\alpha'}(B_r)} \leq A \quad \text{for some } \beta > \alpha',$$

*then  $|u(x)| \leq A|x|^{1+\beta}$  for  $x \in B_{1/2}$ .*

*Proof.* Fix  $x \in B_{1/2}$ . Since  $u(0) = 0$ , the mean value theorem gives  $|u(x)| \leq |Du(\xi)| |x|$  for some  $\xi \in B_{|x|}$ . Since  $Du(0) = 0$ ,

$$|Du(\xi)| = |Du(\xi) - Du(0)| \leq [u]_{C^{1+\alpha'}(B_{|x|})} |\xi|^{\alpha'} \leq A |x|^{\beta - \alpha'} |x|^{\alpha'} = A |x|^\beta,$$

and the claim follows. (The argument is operator-free: it uses only the vanishing jet and the seminorm decay.)  $\square$

*Proof strategy (route (b), increments; 2026-06-15).* The local route (a) of [4] (rescale  $v$ , apply the sharp estimate) is *not available* nonlocally, by Remark 7.2: the tail of the rescaled  $v_l$  diverges for every  $\gamma > 0$ . We therefore follow PTU's increment route, made linear by Proposition 7.5.

- (1) *Increment equation.* For a unit vector  $h$  (small), set  $w^h := v(\cdot + h) - v$ . By Proposition 7.5 (with  $(-\Delta_p)^s v = 0$ , so the right-hand side is 0),

$$L_{K_v^h} w^h = 0, \quad K_v^h(x, y) = (p-1) \int_0^1 |(v(x) - v(y)) + t(w^h(x) - w^h(y))|^{p-2} dt.$$

- (2) *Lowered growth and finite tail.* Since  $v \in C^1$  with  $Dv(0) = 0$  and  $|v| \leq C(1 + |x|)^{1+\beta}$ , one has  $|w^h(x)| \leq |h| \sup_{[x, x+h]} |Dv| \leq C_h(1 + |x|)^\beta$  with  $\beta < 1$ ; under (C1) this growth is in the tail space of  $L_{K_v^h}$  (Lemma 7.4).
- (3) *Linear regularity + decay.* On compact sets where  $v$  is nondegenerate,  $K_v^h$  lies in the measurable-kernel ellipticity class of [3] §7, which supplies an interior  $C^{1,\alpha_0}$  estimate with tail term for  $w^h$ . The PTU rescaling/decay argument ( $w_\rho^h := \rho^{-\beta} w^h(\rho \cdot)$ , interior estimate, then  $\|Dw^h\|_{L^\infty(B_{1/2\rho})} \leq C\rho^{\beta-1} \rightarrow 0$  as  $\rho \rightarrow \infty$  because  $\beta < 1$ ) forces  $Dw^h(0) = 0$ ; running it at every centre (where ellipticity is locally uniform — this replaces translation invariance) gives  $Dw^h \equiv 0$ , so  $w^h$  is constant.
- (4) *Reconstruction.*  $w^h \equiv c(h)$  for all small  $h$ ; the cocycle  $w^{h_1+h_2} = w^{h_1}(\cdot + h_2) + w^{h_2}$  forces  $c$  additive and (by continuity) linear,  $c(h) = b \cdot h$ ; but  $w^h(0) = v(h)$  has a double zero at  $h = 0$ , so  $b = 0$  and  $v$  is constant on the lattice of increments, hence (with the vanishing jet)  $v \equiv 0$ .

**TODO:** Two genuine gaps, both to raise with R.T.: (i) **nondegeneracy** — that the blow-up limit  $v_0$  has  $Dv_0 \not\equiv 0$  on open sets, so  $K_{v_0}^h$  is uniformly elliptic on compacta (near the branching point  $v$  is flat, but that is a single point); (ii) the precise **linear nonlocal Liouville** “ $L_K w = 0$ ,

$K$  measurable elliptic, sublinear growth  $\Rightarrow$  constant” — locate the exact reference (Kassmann; or [3] §7 run as a Liouville) with the tail hypothesis matching (C1). Step (3)’s “run at every centre” needs the ellipticity constants locally uniform, which is where (i) enters.  $\square$

**Lemma 7.4** (Resolution of the nonlocal tail gap via increments; 2026-06-15). *Let  $v$  satisfy the hypotheses of Conjecture 7.1, so  $|v(x)| \leq C(1 + |x|)^{1+\beta}$  with  $\beta = \alpha - 1$ . The rescaled  $v_l := v(\cdot)/l^{1+\beta}$  has divergent tail  $\text{Tail}_{p-1,sp}(v_l) = \infty$ , because  $(1 + \beta)(p - 1) = \alpha(p - 1) > sp$ ; thus [3] cannot be applied to  $v_l$  directly. Pass instead to first increments  $w^h := v(\cdot + h) - v(\cdot)$ . Since  $v \in C^1$  with  $Dv(0) = 0$ ,*

$$|w^h(x)| \leq |h| \sup_{[x, x+h]} |Dv| \lesssim |h| (1 + |x|)^\beta,$$

one order lower, and the nonlocal tail of  $w^h$  is finite iff

$$\beta(p - 1) = (\alpha - 1)(p - 1) < sp, \quad \text{i.e. iff the tail condition (C1) } \gamma < \frac{(p-1)^2}{p-1+sp}.$$

Hence the incremental quotient — not  $v_l$  — is the object on which the scaling/decay argument can run.

**Proposition 7.5** (Increments solve a linear nonlocal equation; 2026-06-15). *Let  $v \in C_{loc}^1 \cap L_{loc}^\infty$  and  $h \in \mathbb{R}^n$ , and set  $w^h := v(\cdot + h) - v(\cdot)$ . Then, wherever the principal values exist,*

$$(-\Delta_p)^s[v(\cdot + h)](x) - (-\Delta_p)^s v(x) = L_{K_v^h} w^h(x) := c_{n,s,p} \text{P.V.} \int_{\mathbb{R}^n} (w^h(x) - w^h(y)) K_v^h(x, y) \frac{dy}{|x - y|^{n+sp}},$$

with the symmetric, strictly positive kernel

$$K_v^h(x, y) := (p - 1) \int_0^1 |(v(x) - v(y)) + t(w^h(x) - w^h(y))|^{p-2} dt \geq 0.$$

Consequently, if  $v = v_k$  is a blow-up solution as in Proposition 5.1, then

$$L_{K_{v_k}^h} w_k^h = \theta_k^{\gamma-(p-1)} \left[ (u \mapsto u_+^\gamma - u_-^\gamma)(v_k)(\cdot + h) - (u \mapsto u_+^\gamma - u_-^\gamma)(v_k) \right] =: g_k^h,$$

and  $g_k^h \rightarrow 0$  locally uniformly. So  $w_k^h$  solves a linear nonlocal equation with right-hand side tending to 0 — the structure that makes the datum/truncation argument of (c') available for  $p \neq 2$ .

*Proof.* Translating  $y \mapsto y + h$  in the integral for  $(-\Delta_p)^s[v(\cdot + h)](x)$  writes it as  $c \text{P.V.} \int |(V(x) - V(y))|^{p-2} (V(x) - V(y)) |x - y|^{-n-sp} dy$  with  $V := v(\cdot + h)$ . Subtract  $(-\Delta_p)^s v(x)$  and use, for the scalar map  $J_p(t) = |t|^{p-2} t$  with  $J_p'(t) = (p - 1)|t|^{p-2}$  (continuous for  $p \geq 2$ ), the exact identity

$$J_p(a + \delta) - J_p(a) = \left( (p - 1) \int_0^1 |a + t\delta|^{p-2} dt \right) \delta,$$

applied pointwise with  $a = v(x) - v(y)$  and  $\delta = w^h(x) - w^h(y)$  (so  $a + \delta = V(x) - V(y)$ ). The bracket is  $K_v^h(x, y)$ ; symmetry in  $(x, y)$  and nonnegativity are immediate. The displayed right-hand side for  $w_k^h$  is the increment of  $(-\Delta_p)^s v_k = \theta_k^{\gamma-(p-1)} [(v_k)_+^\gamma - (v_k)_-^\gamma]$  from Proposition 5.1, and the prefactor  $\rightarrow 0$ . (*Identity verified numerically for  $p \in \{2, 2.5, 3, 4\}$ ,  $K_v^h > 0$ .*)  $\square$

**Remark 7.6** (What changes for  $p \neq 2$ ). At  $p = 2$ ,  $w^h$  is again  $s$ -harmonic (linearity) and PTU Theorem 3.1 finishes. For  $p \neq 2$ ,  $w^h$  solves not the homogeneous equation but a *linear* nonlocal equation  $L_{K_v} w^h = 0$  with the  $v$ -dependent kernel  $K_v(x, y) \simeq |v(x) - v(y)|^{p-2} |x - y|^{-n-sp}$  (the linearization object of [3] §7). Two consequences. (i) The  $C^{\alpha_0}/C^{1,\alpha_0}$  estimate one applies is the one for this *measurable-kernel* linear class ([3] §7), not Theorem 1.1 for  $(-\Delta_p)^s w = 0$ ; its tail term is finite exactly under (C1). (ii) **Caveat (degeneracy):**  $K_v$  degenerates where  $v$  is flat

( $|v(x) - v(y)|$  small). At the branching point  $v$  has a vanishing 1-jet, so the kernel is weakest precisely there; one must check the blow-up limit  $v_0$  is not flat on an open set (its gradient does not vanish identically away from 0) so that  $K_{v_0}$  lies in a uniform ellipticity class on the relevant region. This is the genuine technical content of the  $p \neq 2$  Liouville and should be flagged to R.T.

*Remark 7.7.*  $\nu = 2$  is out of reach: quadratic polynomials are not  $(s, p)$ -harmonic for  $p \neq 2$ , so even the statement fails at order two; consistent with the absence of  $C^2$  theory for the (local or nonlocal)  $p$ -Laplacian. State this as a remark in the paper, mirroring PTU Remark 4.1 in reverse.

## 8. TARGET THEOREMS

**Theorem 8.1** (Main growth estimate — target). *Let  $(s, p, \gamma)$  lie in the admissible region of §6 and let  $u \in L^\infty(\mathbb{R}^n)$  solve (3). If  $x_0 \in B_{1/2}$  satisfies  $u(x_0) = |Du(x_0)| = 0$ , then*

$$|u(x)| \leq C \|u\|_{L^\infty(\mathbb{R}^n)} |x - x_0|^{\frac{sp}{p-1-\gamma}}, \quad x \in B_{1/2}(x_0),$$

with  $C = C(n, s, p, \gamma) > 0$ .

*Proof modulo Conjecture 7.1.* We run the device of [4] §6 (the quasilinear form of PTU (4.3)–(4.10)) with our ingredients. *Choice of working exponent.* Pick any  $\alpha'$  with

$$0 < \alpha' < \min\{\alpha_0, \alpha - 1\}, \quad \alpha = \frac{sp}{p-1-\gamma},$$

a nonempty range because  $\alpha > 1$  on the admissible set and  $\alpha_0 > 0$  (Lemma 6.3). *This  $\alpha'$  is generic and sub-sharp; the sharp  $\alpha_0$  is used only for this compactness step, not quantitatively.* We may take  $x_0 = 0$ ,  $\|u\|_{L^\infty(\mathbb{R}^n)} = 1$ , and must show  $|u(x)| \leq C|x|^\alpha$ .

*Contradiction and the  $\theta_k$  device.* If not, there are solutions  $u_k$  of (3) with 0 a branching point,  $[u_k]_{C^{1+\alpha'}(B_{1/2})} \leq 2C$ , and points  $x_k$  with  $|u_k(x_k)| > k|x_k|^\alpha$ . Put

$$\theta_k(r') := \sup_{r' < r < 1/2} r^{1+\alpha'-\alpha} [u_k]_{C^{1+\alpha'}(B_r)}, \quad 1 + \alpha' - \alpha < 0.$$

By Lemma 7.3 in contrapositive form (its hypothesis needs  $\alpha - 1 > \alpha'$ , which holds by the choice of  $\alpha'$ ): were  $\theta_k(0^+) \leq k$ , the vanishing jet would force  $|u_k(x)| \leq k|x|^\alpha$ , contradicting  $x_k$ . Hence  $\lim_{r' \rightarrow 0} \theta_k(r') > k$ , and we may pick  $r_k > 1/k$  with

$$r_k^{1+\alpha'-\alpha} [u_k]_{C^{1+\alpha'}(B_{r_k})} \geq \frac{1}{k} \theta_k(1/k) \geq \frac{1}{2} \theta_k(r_k),$$

so  $r_k \rightarrow 0$ . Set  $v_k(x) := u_k(r_k x) / (\theta_k(r_k) r_k^\alpha)$ . *Seminorm bound.* As in [4] (33), for  $1 \leq R \leq \frac{1}{2r_k}$ ,

$$[v_k]_{C^{1+\alpha'}(B_R)} = \frac{(r_k R)^{1+\alpha'-\alpha} [u_k]_{C^{1+\alpha'}(B_{r_k R})}}{\theta_k(r_k)} R^{\alpha-1-\alpha'} \leq R^{\alpha-1-\alpha'},$$

using  $(r_k R)^{1+\alpha'-\alpha} [u_k]_{C^{1+\alpha'}(B_{r_k R})} \leq \theta_k(r_k R) \leq \theta_k(r_k)$ . *Local growth via the gradient anchor.* With  $\eta \equiv 1$  on  $B_{1/2}$ ,  $\eta = 0$  off  $B_1$ ,  $\int_{B_1} \eta D_e v_k = \int_{B_1} D_e \eta v_k \leq C_n$  gives a point  $z \in B_1$  with  $|Dv_k(z)| \leq C_n$ ; combined with the seminorm bound,  $|Dv_k(x)| \leq C R^{\alpha-1}$  and, since  $v_k(0) = 0$ ,  $|v_k(x)| \leq C|x|^\alpha$  for  $1 \leq |x| \leq \frac{1}{2r_k}$ . *Compactness.* Thus  $v_k$  is locally bounded with uniform  $C^{1+\alpha'}$  bounds, so along a subsequence  $v_k \rightarrow v_0$  in  $C_{loc}^{1+\alpha'}(\mathbb{R}^n)$ , with

$$[v_0]_{C^{1+\alpha'}(B_R)} \leq R^{\alpha-1-\alpha'} \quad (R \geq 1), \quad |v_0(x)| \leq C(1 + |x|)^\alpha, \quad [v_0]_{C^{1+\alpha'}(B_1)} \geq \frac{1}{2}.$$

Since 0 is a branching point of each  $u_k$ ,  $v_0(0) = |Dv_0(0)| = 0$ . By Proposition 5.1,  $(-\Delta_p)^s v_0 = 0$  in  $\mathbb{R}^n$ . Its growth exponent  $\alpha$  satisfies  $\alpha < 2$  (Lemma 6.2, C2), i.e.  $\beta = \alpha - 1 < 1$ , so —

provided  $v_0$  is nondegenerate — the hypothesis of Conjecture 7.1 holds and  $v_0 \equiv 0$ , contradicting  $[v_0]_{C^{1+\alpha'}(B_1)} \geq \frac{1}{2}$ .  $\square$

*Note: On the exponent bookkeeping (settled 2026-06-15; this note records the final state after two corrections).* (a) The device runs at a generic, sub-sharp  $\alpha' < \min\{\alpha_0, \alpha-1\}$ , not at level  $1 + \alpha_0$ ; so the earlier “ $1 + \alpha_0 < \alpha < 2$  contradicts the conjecture” panic was a misidentification — there is no contradiction, and  $\alpha_0$  enters only for compactness (any  $\alpha_0 > 0$ ). (b) But the opposite over-correction (“so the binding constraint is the sharp  $\alpha < 1 + \alpha_0$ , CP-style”) is also wrong nonlocally: CP’s one-shot rescaling is **blocked by the tail** (Remark 7.2, verified —  $\alpha(p-1) - sp = \frac{\gamma ps}{p-1-\gamma} > 0$  always). The Liouville must go through increments, whose cap is  $\beta < 1$ , i.e.  $\boxed{\alpha < 2}$  (Lemma 6.2, C2). **Net: the binding constraints are C1 (tail) and C2 ( $\alpha < 2$ ), both explicit; no quantitative lower bound on  $\alpha_0$  is needed.** The price moves elsewhere: the increment route needs the **nondegeneracy** of  $v_0$  (Conjecture 7.1, Remark 7.2) and a linear nonlocal Liouville. This supersedes synthesis finding (iii). Update §6 accordingly when the dust settles; for now the corrected story lives here and in Remark 7.2.

**Corollary 8.2** (Sharp growth of the gradient — target). *Under the conditions of Theorem 8.1,  $\sup_{B_r(x_0)} |Du| \leq C r^{\frac{sp}{p-1-\gamma}-1}$  for  $r < 1/2$ .*

**Theorem 8.3** (Two-phase local  $p$ -dead-core — target, new even locally). *Let  $u$  solve  $\Delta_p u = u_+^\gamma - u_-^\gamma$  in  $B_1$  with  $(p, \gamma)$  admissible. At points with  $u(x_0) = |Du(x_0)| = 0$ ,*

$$|u(x)| \leq C |x - x_0|^{\frac{p}{p-1-\gamma}}.$$

*Proof skeleton. TODO:  $s \nearrow 1$  in Theorem 8.1, after checking (i)  $(-\Delta_p)^s u \rightarrow -\Delta_p u$  in the appropriate sense (locate the Bourgain–Brezis–Mironescu-type statement for  $(-\Delta_p)^s$  and fix normalizing constants  $c_{n,s,p}$ ), and (ii) uniformity in  $s$  of every constant in the chain — in particular of the  $C^{1,\alpha_0}$  estimate as  $s \nearrow 1$  (the window  $p < \frac{2}{1-s}$  opens to all  $p \geq 2$ ). The local two-phase result does not appear in da Silva–Rossi–Salort / da Silva–Salort / Teixeira, which are one-phase: highlight this.*  $\square$

**Theorem 8.4** (Liouville with growth — target). *Entire bounded solutions of (3) in  $\mathbb{R}^n$  with  $u(x) = o(|x - x_0|^{\frac{sp}{p-1-\gamma}})$  as  $|x| \rightarrow \infty$ , for some  $x_0 \in \partial\{u > 0\} \cap \{|Du| = 0\}$ , vanish identically.*

*Proof skeleton. TODO: Transplant PTU Theorem 5.3 verbatim: rescale, apply Theorem 8.1, squeeze. Should be a one-page corollary once the main theorem holds.*  $\square$

**Open Question 8.5.** Nondegeneracy:  $\sup_{B_r(x_0)} u_+ \geq c r^{\frac{sp}{p-1-\gamma}}$  at free boundary points? Requires a barrier: compute (or bound)  $(-\Delta_p)^s$  of  $x \mapsto |x|^\alpha$ . Power-function computations for  $(-\Delta_p)^s$  exist in the literature (1-d computations of Brasco et al.;  $d^s$ -barriers of Iannizzotto–Mosconi–Squassina). Separate note.

## 9. READING LIST WITH SHOPPING LISTS

- **PTU** (this paper, arXiv:2504.09557): §4 line-by-line; extract the  $\theta_k$  device and the exponent bookkeeping.
- **Brasco–Lindgren–Schikorra** (arXiv:1711.09835): shopping list — interior  $C^{0,\gamma_0}$  estimate with explicit tail dependence; the form of  $\text{Tail}_{p-1,sp}$ ; stability under limits.
- **Giovagnoli–Jesus–Silvestre** [3] (arXiv:2509.26565): the  $C^{1,\alpha_0}$  paper for Lemma 6.3 — Theorem 1.1 gives interior  $C^{1,\alpha}$  for  $(-\Delta_p)^s u = 0$ ,  $p \in [2, \frac{2}{1-s})$ , with tail term. Shopping list: constants, uniformity in  $s$ ; **check extension to  $(-\Delta_p)^s u = f \in L^\infty$**  (their statement is homogeneous).

- **Correa–dos Prazeres** [4] (arXiv:2504.11370): two-phase quenching-type (Alt–Phillips) problem for the local  $p$ -Laplacian; §6 runs the Liouville + known-regularity strategy in the local quasilinear setting — the closest blueprint for §7, route (a). [Supervisor’s pointer.]
- **Korvenpää–Kuusi–Lindgren**: weak  $\equiv$  viscosity; cite, do not read deeply on the first pass.
- **da Silva–Salort** (arXiv:2501.13063): the local benchmark  $C^{\frac{p}{p-1-q}}$ ; compare hypotheses and the one-phase vs two-phase gap.
- **Teymurazyan**, *The fractional Laplacian: a primer*: background reference; the author is at KAUST — bring §3–§5 of these notes to his office once Lemmas 6.1–6.3 are verified.

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